Full-production discretization P3: positive-time solution-ball bound

TECT verification-first repository · claim A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM

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1. Purpose and scope

This supplement supplies the missing qualitative bridge from P2 solutions to the P3 Fourier discretization. It concerns the canonical variational three-component flow only, on the fixed torus with positive regularisers and $\eta_{\mathrm{shell}} = 0$. For every initial-radius $R > 0$ and every fixed $0 < \tau \le T < \infty$, write

$$\mathcal{B}_R = \{u_0 \in H^2 : \|u_0\|_{H^2} \le R\}.$$

The result is uniform over $u_0 \in \mathcal{B}_R$ and $t \in [\tau, T]$. It does not assign a numerical value to the resulting constant.

1 Statement

Let u be the P2 solution, P_N the real Fourier projector, and R_N^C the hash-pinned collocation residual. There is a finite constant $C(R, \tau, T)$ such that

$$\sup_{u_0 \in \mathcal{B}_R} \sup_{\tau \le t \le T} \|P_N R(u(t)) - R_N^C(P_N u(t))\|_{L^2} \le C(R, \tau, T) N^{-2}. \tag{1}$$

This is an existence-and-rate statement for the canonical positive-time solution ball. It is not an evaluated error bar for a particular solver run.

2 Uniform positive-time regularity

P2 gives a global H^2 solution, a coercive energy bound, and a smooth coefficient map because both regularisers are positive. Continuity of the functional on the initial H^2 ball gives an upper energy bound depending only on R. The P2 coercivity estimate therefore gives a common H^2 bound for all its trajectories. The analytic-semigroup estimate, the local $H^2 \to L^2$ Moser bound, endpoint Duhamel cancellation, and the two-derivative bootstrap can consequently be applied with constants depending only on (R, τ, T). In particular,

$$\sup_{u_0 \in \mathcal{B}_R,\ \tau \le t \le T} \|u(t)\|_{H^6} \le B_6(R, \tau, T) < \infty. \tag{2}$$

The crucial restriction $\tau > 0$ is explicit: P2 starts at H^2, not at a uniform H^6 ball at time zero.

3 Projection and aliasing estimate

The full residual has principal order four and, by the P2 Moser map, maps bounded H^6 balls to bounded H^2 balls. Fourier projection gives

$$\|(I - P_N)u\|_{H^2} \le C N^{-4} \|u\|_{H^6}. \tag{3}$$

For $f \in H^q(\mathbb{T}^3)$ with $q > 3/2$, the periodic sampling/interpolation aliasing operator obeys

$$\|P_N(I_N - I)f\|_{L^2} \le C_q N^{-q} \|f\|_{H^q}. \tag{4}$$

Choose $q = 2$. It is above the three-dimensional summability threshold, and the $H^6 \to H^2$ residual bound makes (4) applicable to the nonlinear residual. The $H^2 \to L^2$ Lipschitz estimate controls $R(P_N u) - R(u)$ using (3). The collocation residual is exactly the sampled, projected residual of $P_N u$ in the fixed discrete real pairing. Combining these three estimates yields (1); the slower aliasing term is N^{-2}.

4 Numerical verification

The standalone order-accounting audit reads the P1, P2, and P3 source hashes, checks the positive-time H^6 to residual-H^2 order drop, verifies $2 > 3/2$, and evaluates the decreasing Fourier-cutoff envelopes on N8, N12, and N16. It passes 13/13. These checks guard the exponent, torus convention, and source identity; they do not substitute for the analytic Moser proof or produce $C(R, \tau, T)$.

5 Devil's-advocate and mandatory adversarial review

Sign and factor. The residual is the real-L^2 gradient fixed by P1; the estimate compares it only after the same real projection and does not alter its sign. **Convention.** R_N^C remains collocation, not exact Galerkin; the aliasing operator in (4) is the explicit difference. **Units.** The Fourier cutoff is computed from the fixed torus periods, so the Sobolev factors have the correct inverse-length power. **Convergence.** The proof gives an algebraic N^{-2} envelope, while the earlier analytic-field tests may converge faster and must not replace it. **Hardcode masking.** All production coefficients and periods are read from the pinned manifests; only Sobolev orders and grid choices are declared audit inputs. **Limit cases.** At $t = 0$ the uniform H^6 envelope is not available; removal of the rho floor, nonzero shell bias, infinite volume, or the historical backend all fall outside this result. External review of the sampling aliasing lemma and the uniformisation of the P2 bootstrap is invited.

6 What remains open

Equation (1) is qualitative because P2 proves finite smoothing constants but does not yet enclose $B_6(R, \tau, T)$ numerically. Before a Sector-B solver output receives a controlled continuum-error bar, the package must derive an explicit computable enclosure for that constant, propagate it through a dealiased finite-time evolution estimate, and pass independent reproduction.

Result footer

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Result ID:               A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
Precise statement:       Positive-time P2 H2 solution balls obey a qualitative
                         collocation/Galerkin residual bound C(R,tau,T) N^-2.
Scope:                   Fixed T3, canonical P1/P2 functional, eta_shell=0,
                         arbitrary H2 radius R and 0<tau<=t<=T.
Dependencies:            A1 production functional; A2 T6 well-posedness.
Evidence grade:          ANALYTIC + EXECUTED + CONDITIONAL; T3 partial stage.
Reproduction command:    python codes/foundations/
                         a3_full_production_solution_ball_bound.py
Expected output:         13/13 PASS; A3-FULL-SOLUTION-BALL-BOUND-PASS.
Falsification gate:      A3-FULL-DISCRETIZATION-CLOSURE.
Tier before / after:     T3 / T3.
No-overclaim statement:  Not a numerical C(R,tau,T) enclosure, historical
                         solver certificate, t=0 rough-data result, T5/T6/T7.
Next required action:    Enclose B6 and C(R,tau,T), then dealiased evolution.
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